PREOPERATIVE DIAGNOSES:,1. Nasal septal deviation.,2. Bilateral internal nasal valve collapse.,3. Bilateral external nasal valve collapse.

POSTOPERATIVE DIAGNOSES:,1. Nasal septal deviation.,2. Bilateral internal nasal valve collapse.,3. Bilateral external nasal valve collapse.

PROCEDURES:,1. Revision septoplasty.,2. Repair of internal nasal valve collapse using auricular cartilage.,3. Repair of bilateral external nasal valve collapse using auricular cartilage.,4. Harvest of right auricular cartilage.

ANESTHESIA: General endotracheal anesthesia.

ESTIMATED BLOOD LOSS: Approximately 20 mL.

IV FLUIDS: Include a liter of crystalloid fluid.

URINE OUTPUT: None.

FINDINGS: Include that of significantly deviated septum with postoperative changes and a significant septal spur along the floor. There is also evidence of bilateral internal as well as external nasal valve collapse.

INDICATIONS: ,The patient is a pleasant 49-year-old gentleman who had undergone a previous septorhinoplasty after significant trauma in his 20s. He now presents with significant upper airway resistance and nasal obstruction and is unable to tolerate a CPAP machine. Therefore, for repair of the above-mentioned deformities including the internal and external nasal valve collapse as well as straightening of the deviated septum, the risks and benefits of the procedure were discussed with him included but not limited to bleeding, infection, septal perforation, need for further surgeries, external deformity, and he desired to proceed with surgery.

DESCRIPTION OF THE PROCEDURE IN DETAIL: ,The patient was taken to the operating room and laid supine upon the OR table. After the induction of general endotracheal anesthesia, the nose was decongested using Afrin-soaked pledgets followed by the injection of % lidocaine with 1:100,000 epinephrine in the submucoperichondrial planes bilaterally. Examination revealed significant deviation of the nasal septum and the bony cartilaginous junction as well as the large septal spur along the floor. The caudal septum appeared to be now in adequate position. There was evidence that there had been a previous caudal septal graft on the right nares and it was decided to leave this in place. Following the evaluation of the nose, a hemitransfixion incision was made on the left revealing a large septal spur consisting primarily down on the floor of the left nostril creating nearly a picture of the vestibular stenosis on the side. Very carefully, the mucoperichondrial flaps were elevated over this, and it was excised using an osteotome taking care to preserve the 1.5 cm dorsal and caudal strap of the nasal septum and keep it attached to the nasal spine. Very carefully, the bony cartilaginous junction was identified and a small piece of the bone, where the spur was, was carefully removed. Following this, it was noted that the cartilaginous region was satisfactory in quantity as well as quality to perform adequate grafting procedures. Therefore, attention was turned to harvesting the right-sided auricular cartilage, which was

done after the region had adequately been prepped and draped in a sterile fashion. Postauricular incision using a #15 blade, the area of the submucoperichondrial plane was elevated in order to preserve the nice lining and identifiable portion of the cartilage taking care to preserve the ridge of the helix at all times. This was very carefully harvested. This area had been injected previously with 1% lidocaine and 1:100,000 epinephrine. Following this, the cartilage was removed. It was placed in saline, noted to be fashioned in the bilateral spreader graft and alar rim graft as well as a small piece of crush which was used to be placed along the top of the dorsal irregularity. The spreader grafts were sutured in place using submucoperichondrial pockets. After an external septorhinoplasty approach had been performed and reflection of the skin and soft tissue envelope had been performed, adequately revealing straight septum with significant narrowing with what appeared to be detached perhaps from his ipsilateral cartilages rather from his previous surgery. These were secured in place in the pockets using a 5-0 PDS suture in a mattress fashion in two places. Following this, attention was turned to placing the alar rim grafts where pockets were created along the caudal aspect of the lower lateral cartilage and just along the alar margin. Subsequently, the alar rim grafts were placed and extended all the way to the piriform aperture. This was sutured in place using a 5-0 self-absorbing gut suture. The lower lateral cartilage has had some inherent asymmetry. This may have been related to his previous surgery with some asymmetry of the dome; however, this was left in place as he did not desire any changes in the tip region, and there was adequate support. An endodermal suture was placed just to reenforce the region using a 5-0 PDS suture. Following all this, the area was closed using a mattress 4-0 plain gut on a Keith needle followed by the application of *** 5-0 fast-absorbing gut to close the hemitransfixion incision. Very carefully, the skin and subcutaneous tissue envelopes were reflected. The curvilinear incision was closed using a Vicryl followed by interrupted 6-0 Prolene sutures. The marginal incisions were then closed using 5-0 fast-absorbing gut. Doyle splints were placed and secured down using a nylon suture. They had ointment also placed on them. Following this, nasopharynx was suctioned. There were no further abnormalities noted and everything appeared to be in nice position. Therefore, an external splint was placed after the application of Steri-Strips. The patient tolerated the procedure well. He was awakened in the operating room. He was extubated and taken to the recovery room in stable condition.**